NCI

Coláiste Náisiúnta na hÉireann | National College of Ireland

This certificate is awarded to:

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Data Structures and Algorithms CA1 Report

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Github Repository Link: https://github.com/Wrona-03/Data_Structures_CA1.git

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This Java application revolves around the theme of "Urban Greening/Biodiversity" in the Smart Docklands link provided by the lecturer. The problem this app aims to cover is the lack of greenery and flora in the Docklands area. As a heavily urbanised area of Dublin, it features lots of concrete and office buildings but limited biodiversity and green spaces. Smart Docklands already aims to resolve this issue by working with the citizens living and working there. However, a challenge remains: how can public suggestions be easily collected regarding new biodiversity improvements?

This is the problem this app aims to solve. The Java application enables users to select an area within the Dublin Docklands where they would like to see increased biodiversity. Users can then choose from a variety of native Irish flowers, such as primrose, cowslip, and bluebells, and submit a suggestion for planting in that area. This provides members of the public with an accessible and straightforward way to suggest improvements in biodiversity in the area.

The application demonstrates the use of a few abstract data types covered in the module, including queues, stacks, and ArrayLists. The ArrayList holds all the plant suggestions submitted by users. A queue structure represents submissions that are currently pending review, while a stack is used to depict approved submissions. The system operates on a FIFO (First in, first out) algorithm, where the status of the submission depends on their status. The user also has the option to remove their submission, should they change their mind on the area or type of flower.

The GUI (Graphical User Interface) is built using Java Swing in Apache Netbeans. With combo boxes, the application allows users to easily select both a location and a plant type with ease. This design helps keep the system accessible and promotes user participation.

Although the application currently stores information only in memory, it demonstrates the idea of a citizen engagement application for biodiversity planning. In a real-world context, the system could be extended to include file storage or a database, enabling submissions to be kept permanently. Additional features might also encompass more plant species and maps of planting regions, as well as images of the area(s) and flora offered, to provide a visual reference to users.